

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 2

Error Analysis Task

COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 30%

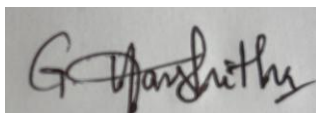
Duration: 30 minutes

Marks: 25

Code of Conduct:

I (G.Harshitha / Reg No:192372384) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



QUESTIONS:

Total Marks:25

1. Error Analysis in Maximum Likelihood Estimation (MLE)

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

Tasks:

- Identify the conceptual error in the likelihood formulation.
- Explain how this mistake affects parameter estimation.
- Suggest the correct mathematical formulation and reasoning.
- How would using log-likelihood fix computational issues?

Answer:

a) Conceptual Error

The student incorrectly writes the likelihood as the **sum of probabilities** instead of the **product of probabilities**. MLE requires multiplying the probability of every independent observation.

b) Effect on Parameter Estimation

- Produces incorrect parameter values.
- Does not maximize the true likelihood.
- Model accuracy decreases even with large datasets.

c) Correct Mathematical Formulation

For independent samples:

$$L(\theta) = \prod_{i=1}^n P(x_i | \theta)$$

Log-likelihood:

$$\log L(\theta) = \sum_{i=1}^n \log P(x_i | \theta)$$

This correctly estimates the parameters by maximizing the probability of observing the data.

d) Why Log-Likelihood?

- Converts multiplication into addition.
- Prevents numerical underflow.
- Makes differentiation and optimization easier.

Conclusion:

Correct likelihood formulation ensures accurate parameter estimation and stable optimization.

2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.

Tasks:

- List possible implementation errors in the learning algorithm.
- Analyze how incorrect learning rate or weight updates affect convergence.
- Identify dataset-related issues that may cause this problem.
- Propose modifications to ensure convergence.

Answer:

a) Possible Implementation Errors

- Incorrect weight update formula.
- Wrong bias update.
- Incorrect prediction rule.
- Improper stopping condition.

b) Effect of Learning Rate

- Very high learning rate causes oscillation.
- Very low learning rate makes convergence slow.
- Incorrect weight updates prevent finding the decision boundary.

c) Dataset Issues

- Incorrect labels.
- Unnormalized features.
- Duplicate or noisy samples.
- Data may not actually be linearly separable.

d) Solutions

- Normalize the data.
- Verify labels.
- Use proper learning rate.
- Correct weight update equation.
- Shuffle training samples.

Conclusion:

Correct implementation and clean data help the Perceptron converge successfully.

3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

Tasks:

- Identify possible mathematical or coding errors in gradient computation.
- Explain the role of activation functions in gradient flow.
- Analyze how vanishing or exploding gradients affect training.
- Suggest techniques to fix the issue (e.g., normalization, initialization).

Answer:

a) Possible Errors

- Wrong derivative calculation.
- Incorrect chain rule implementation.
- Wrong learning rate.
- Coding mistakes in weight updates.

b) Role of Activation Functions

Activation functions control gradient flow.

- Sigmoid may produce very small gradients.
- ReLU reduces the vanishing gradient problem.

c) Vanishing and Exploding Gradients

- **Vanishing gradients:** Weights update very slowly.
- **Exploding gradients:** Weights become extremely large, causing unstable training.

d) Solutions

- Use ReLU activation.
- Apply Batch Normalization.
- Use Xavier or He initialization.
- Apply Gradient Clipping.
- Choose an appropriate learning rate.

Conclusion:

Proper gradient computation and initialization improve stable learning.

4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

Tasks:

- a) Explain why batch gradient descent is slow in large datasets.
- b) Analyze the cause of fluctuations in SGD.
- c) Compare the error behavior of Mini-Batch Gradient Descent.
- d) Recommend the best approach and justify with error trade-offs.

Answer:

a) Why Batch Gradient Descent is Slow

- Uses the entire dataset in every update.
- High computational cost.
- Slow for large datasets.

b) Why SGD Fluctuates

- Updates weights after each sample.
- Gradient estimates are noisy.
- Loss function fluctuates during training.

c) Mini-Batch Gradient Descent

- Uses small batches (32–128 samples).
- Faster than Batch GD.
- More stable than SGD.
- Balances speed and accuracy.

d) Best Recommendation

Mini-Batch Gradient Descent is preferred because:

- Faster convergence.
- Lower computation cost.
- Stable learning.
- Better generalization.

Conclusion:

Mini-Batch Gradient Descent provides the best balance between speed and stability.

5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.

Tasks:

- a) Explain how high dimensionality increases model error.
- b) Identify signs of overfitting in this scenario.
- c) Analyze why distance-based learning fails in high dimensions.
- d) Suggest dimensionality reduction or regularization techniques to improve performance.

Answer:

a) How High Dimensionality Increases Error

- Data becomes sparse.
- Model learns noise instead of patterns.
- Complexity increases.

b) Signs of Overfitting

- High training accuracy.
- Low test accuracy.
- Large gap between training and testing performance.

c) Why Distance-Based Learning Fails

In high dimensions:

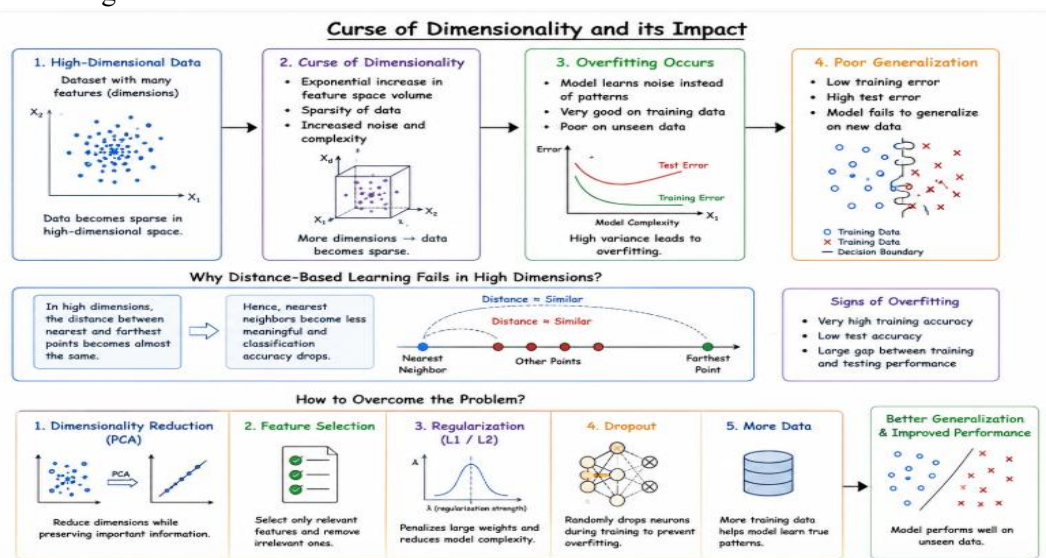
- Distances between points become similar.
- Nearest neighbors become less meaningful.
- Classification accuracy decreases.

d) Solutions

- Apply PCA for dimensionality reduction.
- Use Feature Selection.
- Apply L1/L2 Regularization.
- Use Dropout.
- Collect more training data.

Conclusion:

Reducing dimensions and applying regularization improve model generalization and reduce overfitting.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand